

Digital Concentration Readout of Single Enzyme Molecules Using Femtoliter Arrays and Poisson Statistics

David M. Rissin and David R. Walt*

Department of Chemistry, Tufts University, 62 Talbot Avenue,
Medford, Massachusetts 02155

Received January 31, 2006

ABSTRACT

Methods for accurately quantifying the concentration of a particular analyte in solution are all based on ensemble responses in which many analyte molecules give rise to the measured signal. In this paper, single molecules of β -galactosidase were monitored using a 1 mm diameter fiber optic bundle with 2.4×10^5 individually sealed, femtoliter microwell reactors. By observation of the buildup of fluorescent products from single enzyme molecule catalysis over the array of reaction vessels and by application of a Poisson statistical analysis, a digital concentration readout was obtained. This approach should prove useful for single molecule enzymology and ultrasensitive bioassays. More generally, the ability to determine concentration by counting individual molecules offers a new approach to analysis of dilute solutions.

Introduction. Traditional methods for concentration measurement are derived from a bulk solution response. In this paper, we present a novel method for concentration determination based on Poisson statistics. The bulk solution enzyme concentration is determined by distributing the enzyme-containing solution and a suitable substrate into many femtoliter-sized reactor volumes. The reactor volumes are small enough such that they contain either zero or one enzyme molecule. By observation of the presence or absence of a fluorescent product resulting from single enzyme molecule catalysis in each reaction vessel, a binary readout method can be used to count enzyme molecules. The percentage of reaction vessels occupied by enzyme molecules is correlated to the bulk enzyme concentration. Enzyme catalysis provides amplification and allows us to perform these measurements without the need for true single molecule detection.

A useful approach to isolate and exploit single enzyme molecules is to perform assays in extremely small containers.^{1–9} Micro- and nanotechnologies allow for the design and implementation of small arrayed structures,¹⁰ which are ideal for single molecule isolation and high throughput analysis.⁹ In addition, advances in the sensitivity of analytical methods have resulted in the ability to observe single molecules,¹⁰ revealing new information concerning molecular dynamics and rates of enzymatic catalysis.^{11,12} Other strategies have been employed to confine ultrasmall samples, including, but not limited to, liposomes,^{4,7} oil-dispersed droplets,^{3,8} and chip-based methods.^{2,5} Variation in the reaction

characteristics of single enzyme molecules has been interrogated using a relatively small number of femtoliter-sized reaction vessels¹³ as well as capillaries.¹⁴ Implementing this single enzyme molecule capture technology^{9,13} with a high-density femtoliter array, our objective is to exploit the amplification capabilities of single enzyme molecules across a large number of vessels for the digital readout of enzyme concentration. We employ a prefabricated fiber optic bundle to create an array of individual reaction vessels. This array format is advantageous as it circumvents a complicated microfabrication procedure and provides the ability to observe many reaction vessels simultaneously.

Materials and Methods. Materials. Bundled 4.5 μm optical fibers (1 mm) were purchased from Illumina (San Diego, CA). β -Galactosidase and $\text{Ru}(\text{bpy})_3\text{Cl}_2$ were obtained from Sigma-Aldrich (St. Louis, MO). Resorufin-D- β -galactopyranoside was purchased from Molecular Probes (Eugene, OR). Nonreinforced gloss silicone sheeting material (0.01 in.) was purchased from Specialty Manufacturing Inc. (Saginaw, MI). All other chemicals used were of reagent grade and obtained from Sigma-Aldrich.

Imaging System. A custom-built, upright epifluorescence imaging system acquired all fluorescence images using a mercury light source, excitation and emission filter wheels, microscope objectives, and a CCD camera (QE, Sensicam). Filter wheels and shutters were computer controlled, and analysis was performed with IPlab software (Scanalytics, Fairfax, VA). The system was equipped with a fastening device to fix the fiber optic array onto the system through the entire experiment. A mechanical platform beneath the

* Corresponding author. E-mail: david.walt@tufts.edu.

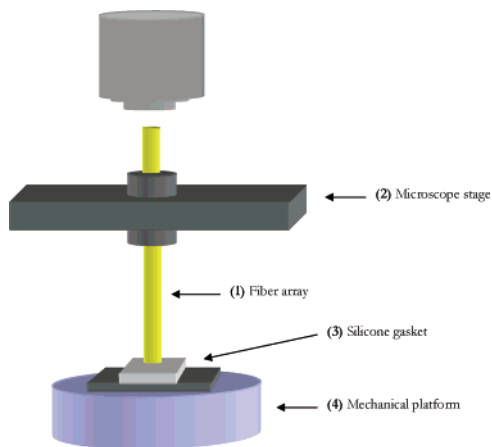


Figure 1. A simple schematic of the femtoliter array and microscope platform. The fiber array (1) is locked into the microscope stage (2). A drop of enzyme/substrate solution is placed on the silicone gasket (3), and the mechanical platform (4) applies pressure to the distal end of the fiber, thereby sealing the solution in each femtoliter chamber.

stage was used to house the silicone-sealing layer, which was subsequently brought into contact with the distal end of the fiber array, sealing off each reaction vessel (Figure 1). All measurements were performed with femtowells arrays at the distal end of the optical fiber bundle.

Integrity of the Seal. A solution of 1 mM Ru(bpy)₃Cl₂ in deionized (DI) water was used for the photobleaching experiments. A piece of silicone, approximately 1 cm², and a microscope slide were cleaned with absolute ethanol using lint-free swabs. The silicone sheeting was placed on the surface of the glass, to which it adhered. Fifty microliters of the Ru(bpy)₃Cl₂ solution was placed on the silicone, and subsequently brought into contact with the fiber bundle, to enclose the solution in the individual vessels. With a field stop on the imaging system, UV light was used to illuminate a small portion of the array for 10 min, photobleaching the Ru(bpy)₃Cl₂. The field stop was then opened, and an image was acquired, displaying the difference in fluorescence. The array was then allowed to rest with the seal maintained. A final image was taken after 60 min, confirming the integrity of the seal.

Enzyme Assays. For the β -galactosidase assay, the substrate used was resorufin- β -D-galactopyranoside. After the individual wells in the array were sealed in the presence of enzyme and substrate, the fluorescence intensity was monitored across the array of vessels for the enzymatic product, resorufin (558 nm excitation/573 nm emission). A 100 μ M solution of resorufin- β -D-galactopyranoside (RDG) was prepared in 100 mM Tris buffer pH 8.0 containing 2.0 mM KCl and 0.1 mM MgCl₂. All enzyme solutions were prepared from previously aliquoted and frozen stock samples in the same reaction buffer. Just prior to experimentation, the two samples were centrifuged for 2 min at 7000 rpm to remove any particulate material that could interfere with the mechanics of the silicone seal. Approximately 1 cm² of silicone and a microscope slide were cleaned with absolute ethanol. The silicone sheeting was placed on the surface of the glass, to which it adhered. Subsequently, 75 μ L volumes of enzyme

and RDG solutions were mixed on the silicone gasket using a pipet. The gasket was mechanically raised toward the distal end of the fiber bundle until it experienced resistance, suggesting that a seal was formed. An initial fluorescence image was acquired, followed by periodic image acquisition for approximately 2 h.

Results and Discussion. Typically, the properties of individual enzyme molecules are hidden in the average bulk solution measurement. The use of small reaction vessels, however, enables individual enzyme molecule responses to be obtained.^{1,9,11–14} A single enzyme molecule can generate a high local concentration of product provided the enzyme and an excess of substrate are contained within a small reaction volume. In the case of femtoliter vessels, a small number of product molecules are sufficient to yield a detectable fluorescence signal. By observing the occupancy of single molecules of β -galactosidase in a large number of wells, it is possible to literally count molecules and determine the concentration of enzyme using Poisson statistics to correlate the number of “active wells” to the bulk solution concentration.

The number and volume of reaction vessels employed govern the dynamic range of concentrations that can be determined by this technique. The reaction vessel volumes employed here are 46 fL (vide infra); therefore, a solution of 3.6×10^{-11} M β -galactosidase will yield, on average, one enzyme molecule per vessel. It is important to note that distributing a solution containing 3.6×10^{-11} M enzyme into an array of reaction vessels will not result in the distribution of exactly one enzyme molecule per vessel; statistically, some vessels will have multiple molecules while others will have zero. In the case where the number of enzyme molecules per vessel is high, the data can be fit to a Gaussian distribution. As the ratio of enzyme molecules to reaction vessels approaches zero, the Poisson distribution applies. This limiting distribution is used to calculate the probability of rare events occurring in a large number of trials.¹⁵ On the basis of Poisson statistics, for a concentration of 3.6×10^{-11} M, a distribution between zero and five enzyme molecules per container is observed, with the most probable values being zero and one.

$$P_{\mu}(v) = e^{-\mu}(\mu^v/v!) \quad (1)$$

As shown in eq 1, the probability of observing v events is calculated based on the expected average number of events per trial, μ . If the concentrations used are much less than 3.6×10^{-11} M, the expected average becomes exceptionally low, the distribution is narrowed, and the probability of observing anything other than zero or one event per trial is improbable in all experimental cases. At these low concentrations, the relationship between the percentage of active reaction vessels and the bulk enzyme concentration is approximately linear. After waiting for sufficient time to allow enzyme catalysis to occur, individual vessels were interrogated for an on/off response, correlating to each vessel either possessing or lacking enzymatic activity. Utilization of a yes or no response, in conjunction with the high-density

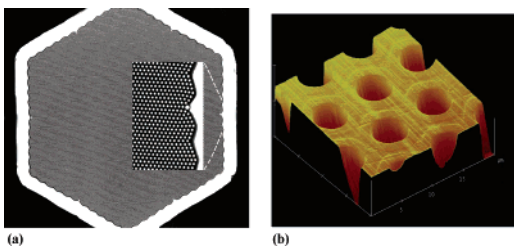


Figure 2. Images of the etched surface of the fiber optic bundles. (a) Entire fiber array and close-up microscope images of the fiber bundle, emphasizing the regularity of both the array and each individual optical fiber. (b) AFM image of a portion of the etched surface, showing wells created from the etching process.

array of reaction vessels, permits the digital readout of bulk concentrations of β -galactosidase. This readout is accomplished by counting the vessels containing an active enzyme molecule across the array, with the resulting “active well” percentage correlating to the enzyme concentration. Due to the large number of vessels simultaneously interrogated on our array (approximately 8000 per experiment), the ratio of enzyme molecules to reaction vessels could be as low as 1:500, as the large number of wells provides a statistically significant signal even at this low ratio.

Optical fiber bundles containing approximately 2.4×10^5 individual $4.5 \mu\text{m}$ diameter optical fibers were used as the substrate for fabricating femtoliter reaction vessel arrays.¹⁶ Fabrication can be performed in a single step, as the optical fiber bundle core material etches at a faster rate than the cladding material (Figure 2). The well volume can be precisely controlled, as etch depth varies with etch time and etchant concentration. The optical fibers used in these experiments were etched to a depth of approximately $2.9 \mu\text{m}$, yielding a 46 fL well volume. The optical fiber bundle is advantageous in that the individual fibers in contact with each well can be used to carry both excitation and emission light via total internal reflection to and from the wells, enabling remote interrogation of the well contents. An array of optical fibers provides the capability for simultaneous excitation of molecules in adjacent vessels, without signal “cross-talk” between fibers.

To seal the femtoliter array, a 0.01-in. thick silicone elastomer gasket was sandwiched between a microscope slide and the fiber array using a mechanical platform. This platform applied uniform pressure to the gasket material, across the entire bundle, sealing off each microwell to create the reaction vessels. The silicone/glass seal used to create and isolate the femtoliter containers was inspected for its sealing ability by performing a photobleaching experiment (Figure 3). This experiment substantiated the integrity of the seal for its ability to successfully isolate the array of vessels. Enzyme molecule denaturation on the glass surface¹⁷ was prevented by blocking with a BSA blocking buffer. Enzyme-to-vessel ratios used ranged from 1:5 down to 1:500, achieving accurate detection over 2 orders of magnitude.

For the β -galactosidase assay, different bulk solution enzyme concentrations correspond to different ratios of enzyme to vessel volume, resulting in variation in the percentage

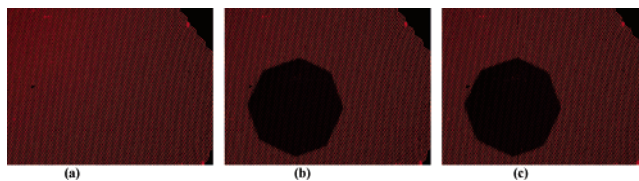


Figure 3. Enclosure into the microchambers and evaluation of the silicone seal for integrity. (a) A solution of $\text{Ru}(\text{bpy})_3\text{Cl}_2$ was enclosed into the array of chambers as observed by the red fluorescence across the array. (b) A small octagonal portion of the fiber bundle was photobleached via UV light. (c) The array was allowed to sit, and a final image was taken 60 min later. Diffusion of $\text{Ru}(\text{bpy})_3\text{Cl}_2$ from one well to another as a result of an imperfect silicone seal would display increased fluorescence intensity in photobleached wells and was not observed. (Pseudocolor added using IPLab software.)

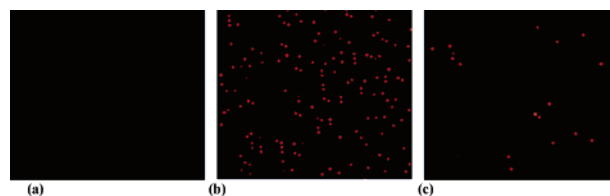


Figure 4. Detection of the activity of single molecules of β -galactosidase: (a) background image of a portion of the array; (b) an image taken of a portion of a 1:5 enzyme to vessel assay; (c) 1:80 enzyme to vessel assay. (Pseudocolor added using IPLab software.)

of vessels that contain an enzyme molecule (Figure 4). Resorufin-D- β -galactopyranoside (RDG) was used as the substrate for the experiments, which was sealed into all the vessels, along with the trapped enzyme molecules, using a silicone gasket material and mechanical arm. The expected percentages of active wells were calculated for each concentration used by applying the Poisson distribution statistics. As seen in the comparison of each experimental result with the percentage of occupied vessels calculated from the Poisson distribution, the array measurements successfully correlated with the number of single enzyme β -galactosidase molecules over the entire range of interrogated concentrations (Table 1). There is disparity in the observed signals as a result of molecule-to-molecule variation in catalytic activity.^{11,13,14,18} This result is most likely due to the inherent stochastic nature of enzymes, in addition to surface effects, resulting in modulation of enzyme activity. Bulk measurements of β -galactosidase using capillary electrophoresis laser-induced fluorescence report a limit of detection (LOD) of 1.5 fM,¹⁹ while our digital readout method LOD is currently capable of detecting β -galactosidase concentrations of 72 fM. Integrating a camera with a higher-density CCD chip should improve the LOD because it will increase the number of individual fibers that are interrogated and will consequently improve the dynamic range of the assay, as the detection limit is reached when only a few vessels on the entire array show enzymatic activity. Similarly, increasing the volume of each reaction vessel by lengthening the fiber etching time will improve the LOD as the probability of isolating a single enzyme molecule in dilute solution increases with increasing vessel volume.

Table 1. Digital Readout from the Arrays^a

digital readout of enzyme concentrations			experiments			actual % active	
enzyme-to-well ratio	concentration	Poisson % active	t1	t2	t3	average	stand dev
1:5	7.20×10^{-12}	18.2	14.87	15.44	14.56	14.96	0.45
1:10	3.60×10^{-12}	9.5	11.46	14.96	6.92	11.11	4.03
1:20	1.80×10^{-12}	4.9	5.57	4.78	5.73	5.36	0.70
1:40	9.00×10^{-13}	2.5	3.47	2.80	1.92	2.73	0.78
1:80	4.50×10^{-13}	1.2	1.46	1.26	1.31	1.34	0.11
1:100	3.60×10^{-13}	1.0	1.10	1.11	1.70	1.30	0.34
1:200	1.80×10^{-13}	0.5	0.25	0.42	0.74	0.47	0.25
1:500	7.20×10^{-14}	0.2	0.10	0.15	0.09	0.11	0.03

^a The actual percentages of chambers exhibiting activity, in comparison to the expected percentage calculated from the Poisson distribution, are listed for the various concentrations analyzed.

The variation between the calculated and experimental results can be attributed to the intrinsic variability associated with the probability distribution, as well as experimental error in the preparation of enzyme solutions. Larger deviation in the results regarding the more concentrated samples may have also been a result of enzyme molecules adhering to the fiber prior to sealing off the wells. Similarly, a negative deviation from the expected average may have been a result of small air pockets remaining in some wells. Air reduces the available volume in the reaction vessel, decreasing the probability of it containing an enzyme molecule. From the three measurements at the lowest concentration (72 fM), 8, 11, and 7 chambers showed enzymatic activity, giving an average response of 8.67 active chambers, with a standard deviation of 2.08. This value is in close agreement with the expected shot noise from this Poisson process (2.94).

Improvements can be made to this technology in the areas of both seal consistency and dynamic range capabilities. Different types of silicone materials are currently being tested for their ability to create a more consistent seal; the silicone gasket material used, however, produced a workable seal over the bundle a majority of the time (>80%). The limitations of this technique are realized both above and below the thresholds of the dynamic range. As the concentration goes below the lower limit of the dynamic range, the number of enzyme molecules is too low to observe sufficient occupied wells, and therefore, the number of wells must be increased in order to make sure that a statistically significant number of them are occupied by enzyme molecules. Results for extremely dilute concentrations have large relative errors associated with them, due to the very small number of reaction vessels that are expected to show activity. Slight deviation from the expected Poisson value, in this case, will result in a large error. The ultimate upper limit to this technique occurs when 100% of the reaction vessels contain at least one enzyme molecule. At this limit, discrimination between two solutions of high enzyme concentrations is not feasible. As the percentage of active vessels approaches 100%, the linearity between concentration and active vessel percentage is lost. This situation results in a broadening distribution, as a normal distribution becomes an increasingly better approximation of the results. This upper limit was not attained in the experiments described here.

Conclusions. Enzymatic reactions in combination with extremely small volumes should have significant application to fundamental enzymology studies, as well as digital concentration measurements. The future of this technology will permit studies with multiple different enzymes and will push the limits of ultralow detection for protein and DNA targets using enzyme-linked detection methods in conjunction with the high-density femtoliter array.

Acknowledgment. The authors thank Dr. David Monk for his help in developing a workable mechanical platform and DARPA for their support of this work.

References

- (1) Lipman, A. E.; Shuler, B.; Bakajin, O.; Eaton, W. A. *Science* **2003**, *301*, 1233–1235.
- (2) Foquet, M.; Korlach, J.; Zipfel, W. R.; Webb, W. W.; Craighead, H. G. *Anal. Chem.* **2004**, *76*, 1618–1626.
- (3) Gratzl, M.; Lu, H.; Matsumoto, T.; Yi, C.; Bright, G. R. *Anal. Chem.* **1999**, *71*, 2751–2756.
- (4) Stamou, D.; Duschl, C.; Delamarche, E.; Vogel, H. *Angew. Chem., Int. Ed.* **2003**, *42*, 5580–5583.
- (5) Nagai, H.; Murakami, Y.; Yokoyama, K.; Tamiya, E. *Biosens. Bioelectron.* **2001**, *16*, 1015–1019.
- (6) Gosalia, D. N.; Diamond, S. L. *Proc. Natl. Acad. Sci. U.S.A.* **2003**, *100*, 8721–8726.
- (7) Chiu, D. T.; Wilson, C. F.; Ryttsen, F.; Stromberg, A.; Farre, C.; Karlsson, A.; Nordholm, S.; Gaggari, A.; Modi, B. P.; Moscho, A.; Garza-Lopez, R. A.; Orwar, O.; Zare, R. N. *Science* **1999**, *283*, 1892–1895.
- (8) Nakano, M.; Komatsu, J.; Matsuura, S.; Takashima, K.; Katsura, S.; Mizuno, A. *J. Biotechnol.* **2003**, *102*, 117–124.
- (9) Rondelez, Y.; Tresset, G.; Tabata, K. V.; Arata, H.; Fujita, H.; Takeuchi, S.; Noji, H. *Nat. Biotechnol.* **2005**, *23*, 361–365.
- (10) Whitesides, G. M. *Nat. Biotechnol.* **2003**, *21*, 1161–1165.
- (11) Xue, Q. F.; Yeung, E. S. *Nature* **1995**, *373*, 681–683.
- (12) Lu, H. P.; Xun, L. Y.; Xie, X. S. *Science* **1998**, *282*, 1877–1882.
- (13) Tan, W. H.; Yeung, E. S. *Anal. Chem.* **1997**, *69*, 4242–4248.
- (14) Craig, D. B.; Arriaga, E. A.; Wong, J. C. Y.; Lu, H.; Dovichi, N. J. *J. Am. Chem. Soc.* **1996**, *118*, 5245–5253.
- (15) Taylor, J. R. *An Introduction to Error Analysis*, 2nd ed.; University Science Books: Sausalito, CA, 1997.
- (16) Pantano, P.; Walt, D. R. *Chem. Mater.* **1996**, *8*, 2832–2835.
- (17) Wheeler, A. R.; Thronset, W. R.; Whelan, R. J.; Leach, A. M.; Zare, R. N.; Liao, Y. H.; Farrell, K.; Manger, I. D.; Daridon, A. *Anal. Chem.* **2003**, *75*, 3581–3586.
- (18) Lee, J. Y.; Li, H. W.; Yeung, E. S. *J. Chromatogr., A* **2004**, *1053*, 173–179.
- (19) Eggertson, M. J.; Craig, D. B. *Biomed. Chromatogr.* **1999**, *13*, 516–519.

NL060227D